ECON2050 Tutorial 4

Basic Real Analysis IV

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Today's Plan

- Remainder: Assignment 1 is due this Friday!
- Tutorial questions: Q1–Q5

Question 1

Determine the first-order Taylor approximation of

$$f: \mathbb{R}^2_{++} \to \mathbb{R}, \qquad f(x,y) = \ln(3x + 2y),$$

around the point

$$\boldsymbol{a} = (2, 1).$$

1:50

$$f(x,y) \approx f(2,1) + \nabla f(2,1) \cdot (x-2, y-1) = \ln 8 + \frac{3}{8}\cdot(x-2) + \frac{2}{8}\cdot(y-1)$$

$$= \ln 8 + \frac{3}{8}x + \frac{1}{4}y - 1$$

$$f(x,y) = \ln(3x+2y)$$

$$f(2,1) = \ln 8$$

$$\frac{\partial f(x,y)}{\partial x} = 3 \cdot \frac{1}{3x+2y}$$

$$\frac{\partial f(x,y)}{\partial y} = 2 \cdot \frac{1}{3x+2y}$$

$$\nabla f(x,y) = \left(\frac{3}{3x+2y}, \frac{2}{3x+2y}\right)$$

$$\nabla f(2,1) = \left(\frac{3}{8}, \frac{2}{8}\right)$$

Tutorial Q1: Knowledge used

Gradient

The gradient of f at $\boldsymbol{x}_0$ is the vector of partial derivatives:

$$\boldsymbol{\nabla} f(\boldsymbol{x}_0) = \left(\frac{\partial f(\boldsymbol{x}_0)}{\partial x_1}, \frac{\partial f(\boldsymbol{x}_0)}{\partial x_2}, \ldots, \frac{\partial f(\boldsymbol{x}_0)}{\partial x_k}\right).$$

First-order approximation

For $\Delta \boldsymbol{x}$ sufficiently close to $\mathbf{0}$,

$$\Delta y = f(\boldsymbol{x}_0 + \Delta \boldsymbol{x}) - f(\boldsymbol{x}_0) \approx \mathrm{d}y = \boldsymbol{\nabla} f(\boldsymbol{x}_0) \cdot \Delta \boldsymbol{x}.$$

Equivalently,

$$f(\boldsymbol{x}) \approx f(\boldsymbol{x}_0) + \boxed{\boldsymbol{\nabla} f(\boldsymbol{x}_0) \cdot (\boldsymbol{x} - \boldsymbol{x}_0)}$$

$f(x): \mathbb{R} \to \mathbb{R}$: $f(x) \approx \underbrace{f(a) + f'(a)(x-a)}_{\text{tangent line}}$

Tutorial Q1: Solution

Solution

First,

$$\nabla f(x, y) = \left(\frac{3}{3x + 2y}, \frac{2}{3x + 2y} \right).$$

At $\boldsymbol{a} = (2, 1)$,

$$f(\boldsymbol{a}) = \ln 8, \qquad \nabla f(\boldsymbol{a}) = \left(\frac{3}{8}, \frac{2}{8} \right).$$

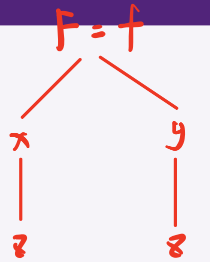
Therefore,

$$\begin{aligned} f(x, y) &\approx \ln 8 + \left(\frac{3}{8}, \frac{2}{8} \right) \cdot (x - 2, y - 1) \\ &= \ln 8 + \frac{3}{8}(x - 2) + \frac{2}{8}(y - 1) \\ &= \ln 8 + \frac{3}{8}x + \frac{1}{4}y - 1. \end{aligned}$$

Question 2

Suppose

$$\boxed{x(z) = z^2 + 1,} \quad \boxed{y(z) = e^z,} \quad f(x, y) = \boxed{\frac{3x^2}{y}}.$$



Let

$$F(z) \equiv f(x(z), y(z)).$$

Calculate $F'(z)$ at $z = 0$.

$$F'(z) = \frac{\partial f}{\partial x} \cdot \frac{dx}{dz} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dz} = \frac{6x}{y} \cdot 2z - \frac{3x^2}{y^2} \cdot e^z$$

$$\frac{\partial f}{\partial x} = \frac{6x}{y}, \quad \frac{dx}{dz} = 2z$$

$$\frac{\partial f}{\partial y} = -\frac{3x^2}{y^2}, \quad \frac{dy}{dz} = e^z$$

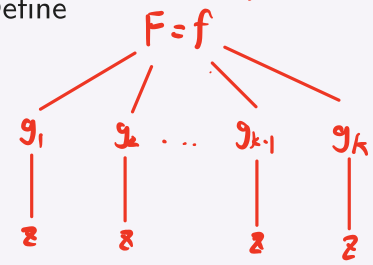
$$= \frac{6 \cdot (z^2+1)}{e^z} \cdot 2z - \frac{3(z^2+1)^2}{e^{2z}} \cdot e^z$$

$$F'(0) = 0 - \frac{3(0+1)^2}{e^0} \cdot e^0 = -3$$

Tutorial Q2: Knowledge used

Generalized chain rule

tree diagram



Suppose $f : \mathbb{R}^k \to \mathbb{R}$ is differentiable and each $g_j : \mathbb{R} \to \mathbb{R}$ is differentiable. Define

$$F(z) = f\big(g_1(z), \ldots, g_k(z)\big).$$

Then

$$F'(z) = \sum_{j=1}^{k} \frac{\partial f}{\partial x_j}\big(g_1(z), \ldots, g_k(z)\big)\, g_j'(z).$$

$$F'(z) = \frac{\partial F}{\partial g_1} \cdot \frac{dg_1}{dz} + \frac{\partial F}{\partial g_2} \cdot \frac{dg_2}{dz} + \ldots + \frac{\partial F}{\partial g_k} \cdot \frac{dg_k}{dz}$$

Interpretation

The total change in the composite function is the sum of the effects through each argument of the outer function.

Tutorial Q2: Knowledge used

Knowledge used

- **Lecture 4:** the generalized chain rule.
- For $F(z) = f(x(z), y(z))$,

$$F'(z) = f_x(x(z), y(z))x'(z) + f_y(x(z), y(z))y'(z).$$

- Differentiate the outer function partially and the inside functions with respect to z.

Tutorial Q2: Solution — derivatives

Solution

We first compute the partial derivatives of the outer function:

$$f_x(x, y) = \frac{6x}{y}, \qquad f_y(x, y) = -\frac{3x^2}{y^2}.$$

For the inside functions,

$$x'(z) = 2z, \qquad y'(z) = e^z.$$

By the generalized chain rule,

$$F'(z) = f_x(x(z), y(z))x'(z) + f_y(x(z), y(z))y'(z).$$

Tutorial Q2: Solution — substitution

Solution

Substituting $x(z) = z^2 + 1$ and $y(z) = e^z$ gives

$$\begin{aligned} F'(z) &= \frac{6(z^2+1)}{e^z}(2z) - \frac{3(z^2+1)^2}{e^{2z}}e^z \\ &= \frac{12z(z^2+1) - 3(z^2+1)^2}{e^z}. \end{aligned}$$

Therefore, at $z = 0$,

$$F'(0) = \frac{12(0)(1) - 3(1)^2}{1} = -3.$$

Tutorial Q3

Question 3

Suppose the Implicit Function Theorem implies that, in a neighborhood of (x_0, y_0, z_0), z is implicitly defined as a function of x and y by

$$ze^x + e^y - ye^z = 0.$$

Compute

$$\frac{\partial z}{\partial x} \quad \text{and} \quad \frac{\partial z}{\partial y}$$

at (x_0, y_0).

Conditions: ① $G(x,y,z) = ze^x + e^y - y \cdot e^z$, $\underline{G \text{ is ctsly differentiable}}$ ✓

all partial derivatives are cts

② $\frac{\partial G}{\partial z}(x_0, y_0, z_0) \neq 0$ ✓

$\frac{\partial G(x,y,z)}{\partial x} = \underset{\text{cts}}{z \cdot e^x}$, $\frac{\partial G(x,y,z)}{\partial y} = \underset{\text{cts}}{e^y - e^z}$, $\frac{\partial G(x,y,z)}{\partial z} = \underset{\text{cts}}{e^x - ye^z} \Rightarrow \frac{\partial G(x_0,y_0,z_0)}{\partial z} = e^{x_0} - y_0 \cdot e^{z_0} \neq 0$

By I.F.T., $\exists$ ctsly differentiable $z = g(x,y)$ satisfying

$$\frac{\partial z}{\partial x} = -\frac{G_x(x_0,y_0)}{G_z(x_0,y_0)}, \quad \frac{\partial z}{\partial y} = -\frac{G_y(x_0,y_0)}{G_z(x_0,y_0)}$$

$$\frac{\partial z}{\partial x} = -\frac{G_x(x_0,y_0)}{G_z(x_0,y_0)} = -\frac{z_0 e^{x_0}}{e^{x_0} - y_0 e^{z_0}}$$

$$\frac{\partial z}{\partial y} = -\frac{G_y(x_0,y_0)}{G_z(x_0,y_0)} = -\frac{e^{y_0} - e^{z_0}}{e^{x_0} - y_0 e^{z_0}}$$

Tutorial Q3: Knowledge used

Implicit Function Theorem

$x = (x_1, \dots, x_m)$

Let $\boldsymbol{x}_0 \in \mathbb{R}^m$, $y_0 \in \mathbb{R}$, and suppose $G(\boldsymbol{x}, y)$ is continuously differentiable around $(\boldsymbol{x}_0, y_0)$. If

$$G(\boldsymbol{x}_0, y_0) = C, \qquad G_y(\boldsymbol{x}_0, y_0) \neq 0,$$

then locally there is a continuously differentiable function $y = g(\boldsymbol{x})$ satisfying

$$G(\boldsymbol{x}, g(\boldsymbol{x})) = C.$$

$$\frac{\partial y}{\partial x_i} = - \frac{G_{x_i}(x_0, y_0)}{G_y(x_0, y_0)}$$

Partial derivatives of the implicit function

For each $i = 1, \dots, m$,

$$\frac{\partial g(\boldsymbol{x}_0)}{\partial x_i} = -\frac{G_{x_i}(\boldsymbol{x}_0, y_0)}{G_y(\boldsymbol{x}_0, y_0)}.$$

Tutorial Q3: Knowledge used

Knowledge used

- **Lecture 4:** the Implicit Function Theorem in the general case.
- Write the equation as $G(x, y, z) = 0$.
- Since z is the implicit variable, the theorem requires

$$G_z(x_0, y_0, z_0) \neq 0.$$

- Then

$$z_x = -\frac{G_x}{G_z}, \qquad z_y = -\frac{G_y}{G_z},$$

  evaluated at (x_0, y_0, z_0).

Tutorial Q3: Solution — setup

Solution

Define
$$G(x, y, z) = ze^x + e^y - ye^z.$$
Then
$$G_x = ze^x, \qquad G_y = e^y - e^z, \qquad G_z = e^x - ye^z.$$
Because the Implicit Function Theorem is assumed to apply at (x_0, y_0, z_0),
$$G_z(x_0, y_0, z_0) = e^{x_0} - y_0 e^{z_0} \neq 0.$$

Tutorial Q3: Solution — implicit derivatives

Solution

By the general Implicit Function Theorem,

$$\frac{\partial z(x_0, y_0)}{\partial x} = -\frac{G_x(x_0, y_0, z_0)}{G_z(x_0, y_0, z_0)} = -\frac{z_0 e^{x_0}}{e^{x_0} - y_0 e^{z_0}}.$$

Similarly,

$$\begin{aligned}\frac{\partial z(x_0, y_0)}{\partial y} &= -\frac{G_y(x_0, y_0, z_0)}{G_z(x_0, y_0, z_0)} \\ &= -\frac{e^{y_0} - e^{z_0}}{e^{x_0} - y_0 e^{z_0}} \\ &= \frac{e^{z_0} - e^{y_0}}{e^{x_0} - y_0 e^{z_0}}.\end{aligned}$$

5 min

Question 4

Obtain an equation of the tangent line to the level curve of $f\colon \mathbb{R}^2 \to \mathbb{R}$ at the given point.

(1) $(x_0, y_0) = (1, 1)$ if

$$f(x, y) = xe^{xy}.$$

(2) $(x_0, y_0) = (-1, 2)$ if

$$f(x, y) = y - x^2.$$

Level curve of level c : $\{(x,y) \in D : f(x,y) = c\}$

Given (x_0, y_0), $f(x_0, y_0) = c$ $\Rightarrow$ tangent line to the level curve of level $f(x_0, y_0)$

$\Leftrightarrow$ tangent line to $\underline{f(x,y) = f(x_0, y_0)}$

equation of x and y.

If $y = g(x)$, tangent line: $y - y_0 = g'(x_0)(x - x_0)$

$\Rightarrow$ Goal: find $g'(x_0) = \frac{dy}{dx}(x_0, y_0)$ $\Rightarrow$ Apply I.F.T

(1) $(x_0, y_0) = (1, 1)$ if

$$f(x, y) = xe^{xy}.$$

(1) $f(x,y) = x \cdot e^{xy}$ $\Rightarrow$ is ctsly differentiable ✓

$\Rightarrow \frac{\partial f(x,y)}{\partial x} = e^{xy} + x \cdot y \cdot e^{xy}$ cts

$\frac{\partial f(x,y)}{\partial y} = x^2 \cdot e^{xy}$ cts

$\frac{\partial f(x_0,y_0)}{\partial y} = \frac{\partial f}{\partial y}(1,1) = 1 \cdot e^{1\cdot 1} = e \neq 0$ ✓

By I.F.T, $\exists$ $y = g(x)$ such that $\frac{\partial y}{\partial x} = -\frac{G_x(1,1)}{G_y(1,1)}$

$\Rightarrow g'(1) = -\frac{e^1 + 1 \cdot 1 \cdot e^1}{1 \cdot e^1} = -\frac{2e}{e} = -2$

$\Rightarrow$ tangent line: $y - 1 = -2 \cdot (x - 1)$

(2) $(x_0, y_0) = (-1, 2)$ if

$$f(x, y) = y - x^2.$$

plug $(x_0,y_0) = (-1,2)$ into f: $f(-1,2) = 2 - 1 = 1$

$\Rightarrow$ $\underline{y - x^2 = f(-1,2) = 1}$

$\frac{\partial y}{\partial x}$ $\Rightarrow$ $y = 1 + x^2$

$\frac{\partial y}{\partial x} = 2x$

$\Rightarrow \frac{\partial y}{\partial x}(-1,2) = -2$

$\Rightarrow$ tangent line: $y - y_0 = \frac{\partial y}{\partial x}(-1,2) \cdot (x - x_0)$

$y - 2 = -2 \cdot (x + 1$

Tutorial Q4: Knowledge used

Implicit Function Theorem

Suppose G is continuously differentiable around (x_0, y_0) and

$$G(x_0, y_0) = C, \qquad G_y(x_0, y_0) \neq 0.$$

Then locally there exists a continuously differentiable function $y = g(x)$ such that

$$G(x, g(x)) = C, \qquad g(x_0) = y_0.$$

Slope of the implicit function

$$g'(x_0) = -\frac{G_x(x_0, y_0)}{G_y(x_0, y_0)}.$$

Tutorial Q4: Knowledge used

Knowledge used

- **Lecture 4:** the Implicit Function Theorem and tangent lines to level curves.
- If the level curve through (x_0, y_0) is locally written as $y = g(x)$ and $f_y(x_0, y_0) \neq 0$, then

$$g'(x_0) = -\frac{f_x(x_0, y_0)}{f_y(x_0, y_0)}.$$

- The tangent line is

$$y = y_0 + g'(x_0)(x - x_0).$$

Tutorial Q4(1): Solution

Solution

For $f(x, y) = xe^{xy}$,

$$f_x = e^{xy}(1 + xy), \qquad f_y = x^2 e^{xy}.$$

At $(1, 1)$,

$$f_y(1, 1) = e \neq 0,$$

so the Implicit Function Theorem applies. Hence

$$g'(1) = -\frac{f_x(1, 1)}{f_y(1, 1)} = -\frac{2e}{e} = -2.$$

Therefore, the tangent line is

$$y = 1 - 2(x - 1) = -2x + 3.$$

Tutorial Q4(2): Solution

Solution

For $f(x, y) = y - x^2$,

$$f_x = -2x, \qquad f_y = 1.$$

At $(-1, 2)$,

$$f_y(-1, 2) = 1 \neq 0,$$

so the Implicit Function Theorem applies. Hence

$$g'(-1) = -\frac{f_x(-1, 2)}{f_y(-1, 2)} = -\frac{2}{1} = -2.$$

Therefore, the tangent line is

$$y = 2 - 2(x + 1) = -2x.$$

Question 5

Calculate the Hessian matrix of each function.

(a)

$$f : \mathbb{R}^2 \to \mathbb{R}, \qquad f(x, y) = xe^{xy}.$$

$$Hf(x) = \begin{pmatrix} 2y \cdot e^{xy} + xy^2 \cdot e^{xy} & 2x \cdot e^{xy} + x^2 \cdot y \cdot e^{xy} \\ 2x \cdot e^{xy} + x^2 \cdot y \cdot e^{xy} & x^3 \cdot e^{xy} \end{pmatrix}$$

(b)

$$f : \mathbb{R}^2_{++} \to \mathbb{R}, \qquad f(x, y) = \ln(xy + y^2 + x^2).$$

(a)

$$f(x,y) = x \cdot e^{xy}$$

$$f_x = e^{xy} + x \cdot y \cdot e^{xy}$$

$$\to f_{xx} = y \cdot e^{xy} + y \cdot e^{xy} + xy^2 \cdot e^{xy} = 2y \cdot e^{xy} + xy^2 \cdot e^{xy}$$

$$\to f_{xy} = x \cdot e^{xy} + x \cdot e^{xy} + x^2 \cdot y \cdot e^{xy} = 2x \cdot e^{xy} + x^2 \cdot y \cdot e^{xy}$$

$$f_y = x^2 \cdot e^{xy}$$

$$\to f_{yx}$$

$$\to f_{yy} = x^2 \cdot x \cdot e^{xy} = x^3 \cdot e^{xy}$$

Tutorial Q5: Knowledge used

Second-order partial derivatives

For a partially differentiable function $f : D \subseteq \mathbb{R}^k \to \mathbb{R}$, the partial derivatives of the first-order partial derivative functions are the **second-order partial derivatives**.

Notation

$$\frac{\partial^2 f}{\partial x_i \partial x_j} \quad \text{or} \quad f_{x_i x_j}$$

denotes the derivative of $\partial f / \partial x_i$ with respect to x_j.
For two variables, the second-order partials are

$$f_{xx}, \qquad f_{xy}, \qquad f_{yx}, \qquad f_{yy}.$$

The derivatives f_{xy} and f_{yx} are called **mixed** or **cross-partial** derivatives.

Tutorial Q5: Knowledge used

Young's Theorem

If $D \subseteq \mathbb{R}^k$ is open and $f : D \to \mathbb{R}$ is twice continuously partially differentiable, then

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}.$$

Thus the order of differentiation does not matter.

$$\begin{tikzcd} & & f_{xx} \\ & f_x \arrow[ur, dash] \arrow[r, dash] & f_{xy} \\ f \arrow[ur, dash] \arrow[dr, dash] & & f_{yx} \\ & f_y \arrow[ur, dash] \arrow[r, dash] & f_{yy} \end{tikzcd}$$

Hessian matrix

The Hessian matrix collects all second-order partial derivatives:

$f : \mathbb{R}^2 \to \mathbb{R}$

$$\boldsymbol{H}_f(\boldsymbol{x}) = \begin{bmatrix} f_{x_1 x_1} & \cdots & f_{x_1 x_k} \\ \vdots & \ddots & \vdots \\ f_{x_k x_1} & \cdots & f_{x_k x_k} \end{bmatrix}.$$

$$Hf(x) = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{xx} \end{pmatrix}$$

If Young's Theorem applies, $\boldsymbol{H}_f(\boldsymbol{x})$ is symmetric.

Tutorial Q5: Knowledge used

Knowledge used

- **Lecture 4:** second-order partial derivatives, Young's Theorem, and the Hessian matrix.
- For a function of two variables,
$$\boldsymbol{H}_f(x, y) = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}.$$
- Compute the gradient first, then differentiate each component again.
- Under the conditions of Young's Theorem, $f_{xy} = f_{yx}$ and the Hessian is symmetric.

Tutorial Q5(a): Solution — first derivatives

Solution

For

$$f(x, y) = xe^{xy},$$

the gradient is

$$\nabla f(x, y) = \left(e^{xy} + xye^{xy},\ x^2 e^{xy}\right).$$

Equivalently,

$$f_x = e^{xy}(1 + xy), \qquad f_y = x^2 e^{xy}.$$

We differentiate these two expressions once more to construct the Hessian.

Tutorial Q5(a): Solution — Hessian

Solution

The second-order partial derivatives are

$$f_{xx} = e^{xy}(2y + xy^2), \qquad f_{xy} = f_{yx} = e^{xy}(2x + x^2 y),$$

and

$$f_{yy} = x^3 e^{xy}.$$

Therefore,

$$\boldsymbol{H}_f(x, y) = e^{xy} \begin{bmatrix} 2y + xy^2 & 2x + x^2 y \\ 2x + x^2 y & x^3 \end{bmatrix}.$$

Tutorial Q5(b): Solution — gradient

Solution

Let
$$s(x, y) = xy + y^2 + x^2.$$
Then
$$f(x, y) = \ln s(x, y),$$
so the gradient is
$$\nabla f(x, y) = \left(\frac{y + 2x}{s(x, y)}, \frac{x + 2y}{s(x, y)} \right).$$
We now differentiate these two components again to obtain the four second-order partial derivatives.

Tutorial Q5(b): Solution — Hessian

Solution

Using the quotient rule,

$$f_{xx} = \frac{y^2 - 2xy - 2x^2}{\left(xy + y^2 + x^2\right)^2},$$

$$f_{xy} = f_{yx} = \frac{-x^2 - y^2 - 4xy}{\left(xy + y^2 + x^2\right)^2},$$

and

$$f_{yy} = \frac{x^2 - 2y^2 - 2xy}{\left(xy + y^2 + x^2\right)^2}.$$

Hence

$$\boldsymbol{H}_f(x, y) = \frac{1}{\left(xy + y^2 + x^2\right)^2} \begin{bmatrix} y^2 - 2xy - 2x^2 & -x^2 - y^2 - 4xy \\ -x^2 - y^2 - 4xy & x^2 - 2y^2 - 2xy \end{bmatrix}.$$